MIL-H-51166A(MU)
26 February 1964
SUPERSEDING
MIL-H-51166(MU)
30 August 1963

MILITARY SPECIFICATION

HOOD, FIELD PROTECTIVE MASK, M6

1. SCOPE

1.1 This specification covers one type and size of protective hood.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

U-T-30 - Talcum; Powdered, Plain and Borated.
CCC-T-191 - Textile Test Methods.
PPP-B-636 - Box, Fiberboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-A-5092 - Adhesive, Rubber (Synthetic and Reclaimed Rubber Base).

FSC 4240

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-175 - Adhesives, Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

DL5-35-66 - Hood, Field Protective Mask, M6.
(All drawings referenced in this specification are listed on Drawing DL5-35-66.)

PUBLICATIONS

QUARTERMASTER CORPS PURCHASE DESCRIPTION

LP/DES 27-61 - Cloth, Coated, Nylon, Butyl Coated, Toxicological Agents Protective.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-35-66 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the hood shall conform to the specifications listed on Drawing DL5-35-66 and on subsidiary drawings.

3.2 Assembly. The hood shall be assembled in accordance with Drawing D5-35-66.

3.2.1 Stitching. All rows of stitching shall be regular lines free from loose stitching or thread loops. All stitching shall be of continuous thread. All seams shall be backstitched at the ends with a minimum of 3 stitches.

3.3 Fastening devices. The pile and hook fastener, drawcord, buckle, drawcord clip, and spring shall function smoothly and efficiently.

3.4 Light penetration. When the hood is examined over a 100-watt light as specified in 4.7.1, there shall be no light penetration through the hood to within 2.0 inches of the bottom, with the exception of the seams around the five openings in the front. If light is visible through six or more holes in the seams or through any hole in the body, a hood is considered defective.

3.4.1 Windows (film transparencies). Windows are acceptable provided that they show no evidence of water leakage when tested as specified in 4.7.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.

3.4.2 Luting compound. A luting compound may be used on the seams of the hood prior to the light penetration test. The luting compound may be applied on the inside or the outside of the hood. When used on the outside, the luting compound shall be applied in a neat manner and shall match the color of the hood. The seams shall be treated with an approved preparation (see 6.6) prior to the application of the luting compound. The luting compound shall be a self-curing solvent neoprene rubber diisocyanate adhesive (see 6.4). When used in combination with the coated fabric, the luting compound shall meet the following requirements when tested as specified in 4.7.3.

Blocking of seams	Scale rating No. 3
Resistance to low temperature	Pass test with no cracking, flaking, or separation from the fabric
Stiffness (centimeters)	
Initial	10 max.
*After heat aging	12 max.
Solids content	20 min.

*The specimen shall show no cracking, flaking or separation from the fabric.

3.4.3 Dusting. After the luted seams have thoroughly dried, they shall be dusted with talcum powder conforming to Specification U-T-30.

3.5 Fit. The hood must assemble to the M17 mask without distortion of the hood or the mask (see 4.7.4.).

3.6 Repairs. Repairs are permitted on the hood for cuts, nicks (1/4-inch maximum length), or pinhole punctures. Repairs shall be accomplished by securely cementing an approximate 1-inch square patch of material conforming to Specification LP/DES 27-61 to the inside of the hood so as to completely cover the defective area. The cement shall conform to type I or type II of Specification MIL-A-5092. Prior to the application of the cement, the surface shall be prepared with Hylene M-50. No more than one repair will be permitted on any single hood and each repaired hood shall be considered a minor defect when examined in accordance with 4.6.4.1.

3.7 Workmanship. The hood shall be free from damage (tear, rip, cut, nick, abrasion, or puncture) and foreign matter (dirt, stain, grease, or oil). All seams shall be free from pleating or distortion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Supplier's responsibility will be as specified in the contract or order.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Initial production. Prior to the start of regular production, the supplier shall submit initial production samples of seams coated with luting compound (4.7.3.1). The samples shall be made of the same materials as will be used in regular production and shall be manufactured with the same equipment and in accordance with the same processes and procedures as will be used in regular production. All materials shall be obtained from the same source of supply. Approval of initial production will be based on whether the samples meet the requirements of 3.4.2. The Government reserves the right to require new initial production until such time as acceptable samples are submitted and approved. New samples shall be submitted whenever a new lot of luting compound is used.

4.4 Lotting.

4.4.1 Hoods. A lot shall consist of the hoods produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions. However, no more than five consecutive lots of cloth from one manufacturer shall be used to make one lot of hoods. To facilitate manufacturing the strap, reinforcement strip, and reinforcement patch, components need not be considered in the application of this lotting criteria.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.

4.4.2 Blanks. A lot shall be that quantity of blanks produced from one cutting.

4.5 Sampling.

4.5.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with Standard MIL-STD-105. For die-cut blanks, Level S-1 shall be used.

4.5.2 For pattern configuration. When blanks are knife-cut, two samples shall be taken from each blank lot (see 4.4.2), one from the top and one from the bottom.

4.5.3 For fit. Five samples shall be taken from each lot of finished hoods.

4.6 Inspection provisions.

4.6.1 For examination and nondestructive tests. Sample hoods and the level A preparation for delivery thereof shall be examined and tested in accordance with the classification of defects and Standard MIL-STD-105.

4.6.2 For pattern configuration. Sample blanks shall be examined for compliance with the applicable pattern. The acceptance number is zero.

4.6.3 For fit. Sample hoods shall be examined for fit in accordance with 4.7.4. The acceptance number is zero.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.

4.6.4 Classification of defects.

4.6.4.1 Hood, Field Protective Mask, M6 (Dwg D5-35-66).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Heavy coated side of cloth not on exterior of hood	
102	Component match marks misaligned by more than 1/8 inch	
103	Distortion or pleating affecting fit	
104	Component missing, incorrect, or incorrectly assembled	
105	Fastening devices (pile and hook fastener, drawcord, buckle, drawcord clip, or spring) not functioning properly	
106	Stitching run off or stitches broken more than 1/2 inch	
107	Number of stitches per inch incorrect (two more or two less than specified)	
108	Dusting of luted seams with talcum powder insufficient to prevent tackiness	
109	Marking incorrect	
110	Damage (tear, rip, cut, nick, puncture, or abrasion) or contamination other than 209	
111	Misalignment of inner lining and hood body by more than 1/8 inch	
112	Overlap of reinforcement strip less than 3/8 inch	
113	Dimensions of inlet valve or lens opening not as specified	4.7.5
114	Pattern configuration incorrect (except knife-cut blanks, see 4.6.2)	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.

4.6.4.1 Hood, Field Protective Mask, M6 (Dwg D5-35-66). (Cont'd)

Categories	Defects	Acceptance standards
Major A:	AQL 0.65 percent defective	
150	Light penetration through hood	4.7.1
Minor:	AQL 6.5 percent defective	
201	Distortion or pleating not affecting fit	
202	Stitching run off or broken more than 1/4 inch but less than 1/2 inch	
203	Stitching tension loose, causing exposed loops; or tight, causing pleating	
204	Stitching margins not as specified	
205	Number of stitches per inch incorrect (one more or one less than specified)	
206	Thread ends not trimmed to 1/4 inch or less	
207	Seam not back-stitched at least three stitches (unless held by another row of stitches)	
208	Stitch type or pattern not as specified	
209	Minor contamination not affecting serviceability (small areas of stain, dirt, or machine lubricant, other than oil or grease, not exceeding 1/4 square inch and a total area of contamination not exceeding 1 square inch)	
210	Marking illegible or improperly located	
211	Overlap of reinforcement strip more than 3/8 inch but less than 1/2 inch	
212	Repaired hoods (see 3.6)	

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:51Z
Check the source to verify that this is the current version before use.

4.6.4.2 Preparation for delivery (see section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Item not dusted with talcum powder
102	Shipping container reinforcement requirements missing or incorrect
103	Marking missing, incorrect, or illegible

4.7 Tests.

4.7.1 Light penetration. Sample hoods shall be examined over the specified light at rated voltage. The light shall be no farther than 12 inches from the area examined and the observer no farther than 24 inches. The seam shall be examined from several angles. The light intensity of the area around the hood shall be no more than 32 foot-candles.

4.7.2 Window hydrostatic test. Test samples of windows shall be subjected to a hydrostatic pressure of two pounds per square inch for one minute.

4.7.3 Luting compound.

4.7.3.1 Samples. Approximately 2 yards of sample seam 6 inches wide shall be prepared by joining two pieces of the coated fabric with the type of stitch and seam defined in Section LL on Drawing D5-35-66. The seam shall be coated with luting compound after it has been treated with an approved preparation.

4.7.3.2 Blocking resistance. Resistance to blocking of the luting compound shall be tested on a sample prepared as specified above. Test method 5872 of Specification CCC-T-191 shall be used with the following exceptions:

(1) Test specimens shall measure approximately 6 by 8 inches with the seam in the center running the full 8-inch length.

(2) The seam shall be folded over on itself once to a size of approximately 6 by 4 inches so that the luting compound is on the inside of the folded specimen.

(3) The test temperature shall be 160° ± 1.6°F.

(4) A 1-pound weight shall be distributed evenly along the seam.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.

4.7.3.3 Low temperature resistance. The resistance of the luting compound to low temperature shall be tested on a sample prepared as specified above. Test method 5874 of Specification CCC-T-191 shall be used with the following exceptions:

(1) Test specimens shall measure approximately 6 by 8 inches with the seam in the center running the full 8-inch length.

(2) The seam shall be folded over on itself once to a size of approximately 6 by 4 inches so that the luting compound is on the inside of the folded specimen.

(3) The roller weight shall be run along the fold line.

4.7.3.4 Stiffness (initial). A two-mil film (dry thickness) of the luting compound, approximately 3 by 12 inches, shall be cast on the coated fabric. The film shall be allowed to air dry for 24 hours. Test specimens shall be cut from the above prepared cloth and the stiffness shall be determined as specified in method 5204 of Specification CCC-T-191.

4.7.3.5 Stiffness (after aging). Test specimens prepared as specified in 4.7.3.4 shall be placed in a circulating air oven at 180° ± 2.0°F for 24 hours. After the specimens are removed from the oven, they shall be allowed to cool a minimum of 1 hour before testing in accordance with method 5204 of Specification CCC-T-191.

4.7.3.6 Solids content. The solids content of the luting compound shall be determined by method 4021 of Federal Standard 175.

4.7.4 Fit (see 6.7).

4.7.4.1 Distortion. The hood shall be assembled to an M17 mask and observed for stressed seams in the hood and for distortion of the hood or mask. (Stretching of the inlet valve or the lens opening of the hood to effect assembly to the mask is not considered distortion nor is puckering or pleating of the hood material if it does not result in stressing of the seams.)

4.7.4.2 Voicemitter opening spring.

4.7.4.2.1 Outlet valve inside hood. The hood shall be assembled to an M17 mask with the outlet valve inside the hood. The spring in the voicemitter opening must not cover any portion of the holes in the round guard of the diaphragm in the voicemitter.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:51Z
Check the source to verify that this is the currentversion before use.

4.7.4.2.2 Outlet valve outside hood. The hood shall be assembled to an M17 mask with the outlet valve outside the hood. The spring in the voicemitter opening must fit the underside of the voicemitter outlet valve assembly snugly.

4.7.5 Inlet valve and lens opening. The inlet valve and lens openings shall be assembled on a split-plug fixture which approximates the configuration of the lens or inlet valve openings as applicable and applies a circumferential tension of 25.0 ± 1.0 pounds to the inlet valve opening or 20.0 ± 1.0 pounds to the lens opening. The inside circumference of the openings shall be determined under these tensions and shall conform to Drawing D5-35-66.

5. PREPARATION FOR DELIVERY (For repair parts packaging see 6.3)

5.1 Packaging.

5.1.1 Level A.

5.1.1.1 Unit packaging. Each hood shall be dusted lightly on both sides with talcum powder conforming to Specification U-T-30 and then shall be folded and packaged in accordance with method IC-1 or IC-3 of Specification MIL-P-116.

5.1.1.2 Intermediate packaging. Ten hoods individually packaged as specified in 5.1.1.1 shall be intermediately packaged in a W6c, style RSC, fiberboard box conforming to Specification PPP-B-636. In addition, the box seams and manufacturer's joint shall be taped as specified for interior boxes with tape conforming to Specification PPP-T-76.

5.1.2 Level C. The hoods shall be packaged in accordance with the manufacturer's commercial practice, suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A. Ten intermediate packages of hoods packaged as specified in 5.1.1 shall be packed in a V3c, style RSC, fiberboard box conforming to Specification PPP-B-636. In addition, the box seams and manufacturer's joint shall be taped as specified for exterior boxes. When metal banding is used, it shall be galvanized.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:51Z
Check the source to verify that this is the current version before use.

5.2.2 Level C. Hoods packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified in the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The hood is intended for use with the M17 protective mask.

6.2. Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of level of packaging and packing.

6.3 Repair parts packaging. When repair parts packaging is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the applicable components of the M6 hood should be used. (Copies of repair parts packaging data sheets may be obtained from the Director of Engineering and Industrial Services, ATTN: Chief, Design Branch, Edgewood Arsenal, Maryland - 21010.)

6.4 Luting compound. The following compound has been used satisfactorily and may be used provided it meets the requirements of 3.4.2.

Part I	*Part II
100.0 parts by weight of Neoprene (type WRT)	10 parts by weight of Hylene M
4.0 parts by weight of calcined MgO (extra light)	
2.0 parts by weight of Neozone D	
5.0 parts by weight of zinc oxide	
145.0 parts by weight of methyl-ethyl-ketone	*Shall be added to Part I just prior to use
75.0 parts by weight of Toluene	
2.5 parts by weight of Thiuram E - plus dye	

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:51Z
Check the source to verify that this is the currentversion before use.

6.5 International interest. Certain provisions of the specification are the subject of ABC international standardization agreements. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform the respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.6 Hylene M-50. Hylene M-50 has been used satisfactorily to prepare the seams for the luting compound.

6.7 M17 masks. The M17 masks required for use in performing the fit test will be furnished by the Government and should be obtained as directed by the contracting officer.

Custodian:

Army - MU

International interest (see section 6)

Preparing activity:

Army - MU

Project No. 4240-A179

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:51Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:51Z
Check the source to verify that this is the current version before use.